Syllabus Year: 2023-2024

CHRIST (Deemed to be University), Bengaluru - 560029

End Semester Examinations - November 2025

Bachelor of Computer Applications SEMESTER - IV

Course Code: BCA431 **Max Marks: 100**

Course Name: GRAPH THEORY **Duration: 03:00**

General Instructions:

- All rough work should be done in the answer script. Do not write or scribble in the question paper except your register number.
- Verify the Course code / course title and number of pages of questions in the question paper.
- Make sure your mobile phone is switched off and placed at the designated place.
- Malpractices will be viewed very seriously.
- Answers should be written on both sides of the paper in the answer book. No sheets should be detached from the answer book.
- Answers without the question numbers clearly indicated will not be valued. No page should be left blank in the middle of the answer book.

Course Outcome: Students will be able to:

CO1: Enhance research, inquiry and analytical thinking abilities.

CO2: To improve proof-writing skills.

CO3: Apply the basics of Graph Theory to solving practical problems.

	Section A		**3*10=30**
1)	Define preorder traversal of an ordered rooted tree. Give example.	**3**	(CO1, L5)
2)	Define preorder traversal of an ordered rooted tree. Give example.	**3**	(CO1, L5)
3) a)	Draw a disconnected simple graph on 6 vertices with maximum number of edges.	**3**	(CO3, L5)
	OR		
b)	Draw a disconnected simple graph on 6 vertices with maximum number of edges.	**3**	(CO3, L5)
4)	What is meant by Collaboration Graph? Give example.	**3**	(CO2, L2)
5)	Draw a disconnected simple graph on 6 vertices with maximum number of edges.	**3**	(CO3, L5)
6)	Define a spanning tree. Find a spanning tree of (i) (ii) . 5 5	**3**	(CO1, L2)
7)	State Euler's formula for planar graphs and verify the same for the following graphs. (i) (ii) . 4 4	**3**	(CO3, L5)
8) a)	Distinguish Eulerian and Hamiltonian graphs. Give an example of a graph that is neither Eulerian nor Hamiltonian.	**3**	(CO1, L4)
	OR		
b)	What is meant by Collaboration Graph? Give example.	**3**	(CO2, L2)
9)	Explain the Chinese postman problem.	**3**	(CO3, L5)
10)	Define a spanning tree. Find a spanning tree of (i) (ii) . 5 5	**3**	(CO1, L2)
	Section B		**7*3=21**
11)	(i) Explain the concepts of (a) subgraph (b) complement of a graph. How many subgraphs with at least one vertex does have? Also draw the complement 3 of . (ii) If it exists, draw a tree whose complement is also a tree. Justify. 5	**7**	(CO2, L2)
12) a)	(i) Explain the concepts of (a) subgraph (b) complement of a graph. How many subgraphs with at least one vertex does have? Also draw the complement 3 of . (ii) If it exists, draw a tree whose complement is also a tree. Justify. 5	**7**	(CO2, L2)

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OR

b)	If is a simple connected planar graph with vertices, edges and r regions, then prove that = − + 2.	**7**	(CO2, L2)
13)	Define (i) cut vertices (ii) cut edges (iii) cut set with appropriate examples.	**7**	(CO2, L4)

Section C **7*4=28**

14)	Discuss on the applications of graph colorings on (i) Scheduling Final Exams and (ii) Index Registers.	**7**	(CO1, L4)
15)	(i) Define (a) -ary tree (b) full -ary tree. Prove that a tree with vertices have −1 edges. (ii) How many pendant vertices will a complete binary tree on vertices have? Justify your answer.	**7**	(CO2, L2)
16) a)	Find the number of paths of length four from the vertex to vertex in a simple graph G given by	**7**	(CO3, L5)

OR

b)	Use Huffman coding to encode the following symbols with frequencies listed below: : 0.20, : 0.10, : 0.15, : 0.25, : 0.30 What is the average number of bits used to encode a character? OR (i) Explain depth-first search algorithm. Use depth-first search to find a spanning tree of . 5 (ii) Is every internal vertex of a tree a cut-vertex of it? Justify your answer	**7**	(CO1, L3)
17)	Discuss on the applications of graph colorings on (i) Scheduling Final Exams and (ii) Index Registers.	**7**	(CO1, L4)

Section D **7*3=21**

18)	Discuss the algorithm for coloring a graph based on vertex degrees and adjacency relationship with example.	**7**	(CO2, L2)
19) a)	Use Huffman coding to encode the following symbols with frequencies listed below. : 0.10, : 0.25, : 0.05, : 0.15, : 0.30, : 0.07, : 0.08. What is the average number of bits used to encode a character?	**7**	(CO3, L4)

OR

b)	Discuss on the applications of graph colorings on (i) Scheduling Final Exams and (ii) Index Registers.	**7**	(CO1, L4)
20)	If G is a simple connected planar graph, then prove that there is a vertex of degree not exceeding five.	**7**	(CO1, L4)

Revised Bloom's Taxonomy (RBT) Levels

L1 - Remember
L2 - Understand
L3 - Apply
L4 - Analyse
L5 - Evaluate
L6 - Create

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